



Automated Functionality Detection System for Outdoor Advertising

BY

W.T. SANGEETH

202152

Internal Supervisor: Mr. P.G.T.R. Payagala

External Supervisor: Eng. A.S. Anupama Silva



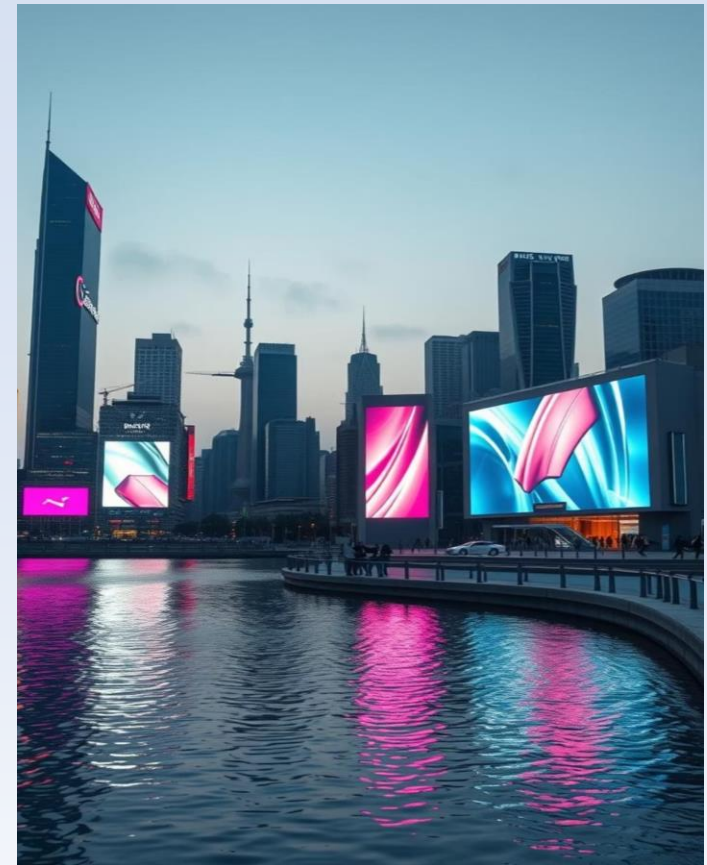
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Introduction

- Outdoor LED advertising screens are everywhere.
- They are expensive and generate revenue only when working.
- Currently, failures are found through manual checks or public reports.
- This leads to long downtime and lost money.





Problem Identification

- **No Real-Time Monitoring:** Operators are blind to failures.
- **Inefficient Maintenance:** Technicians visit without knowing the problem, often requiring multiple trips.
- **Revenue Loss:** Every hour of downtime means lost advertising income.
- **No Fault Details:** Current systems only say "on" or "off," not what is broken.



Research Objectives

- **Primary Objective:**

- Develop an IoT-based system to detect and report LED display faults automatically.

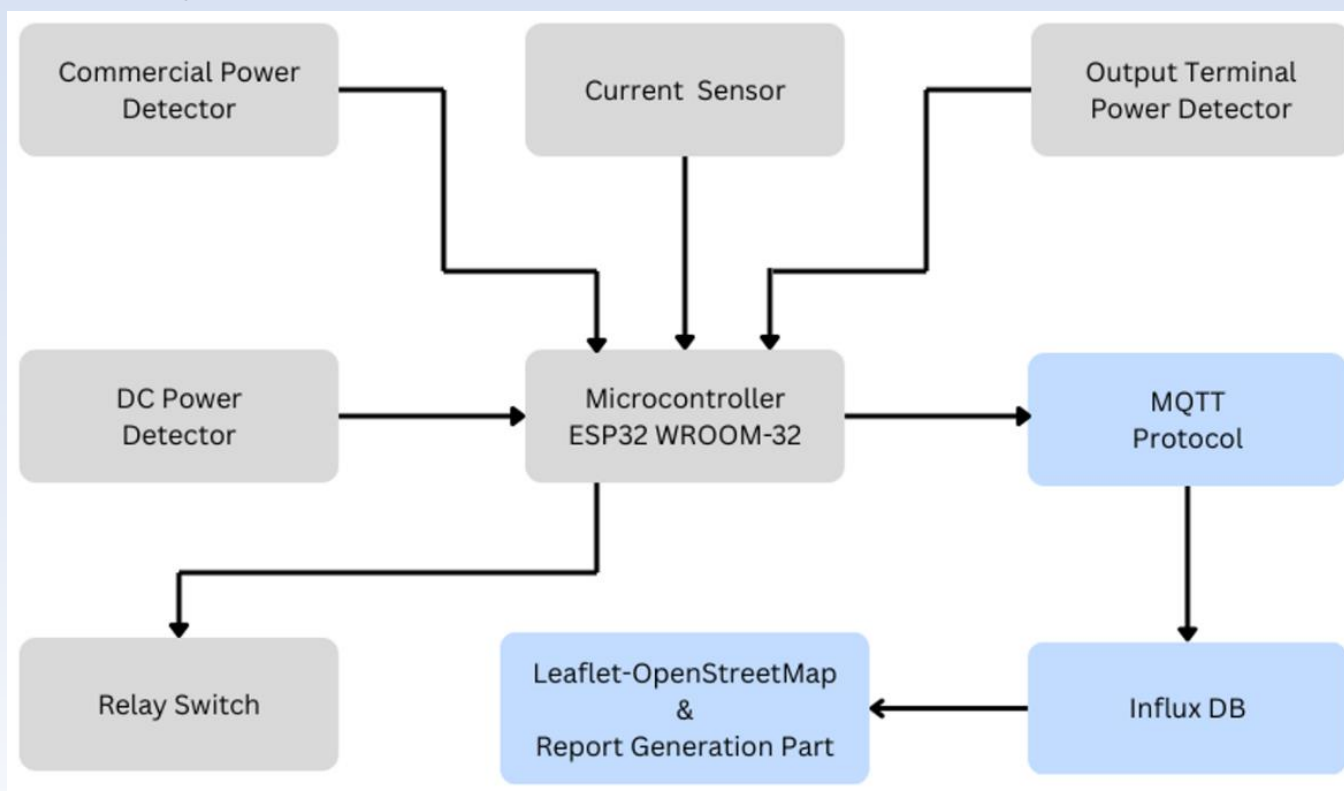
- **Specific Objectives:**

- Classify fault types.
- Send data in real-time to cloud.
- Show results on web dashboard.
- Generate automated PDF/CSV reports.



Methodology

Overall System Architecture

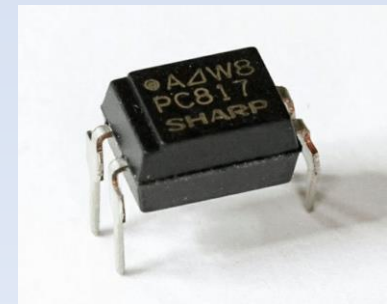




Methodology Cont.

Hardware Design (Sensing)

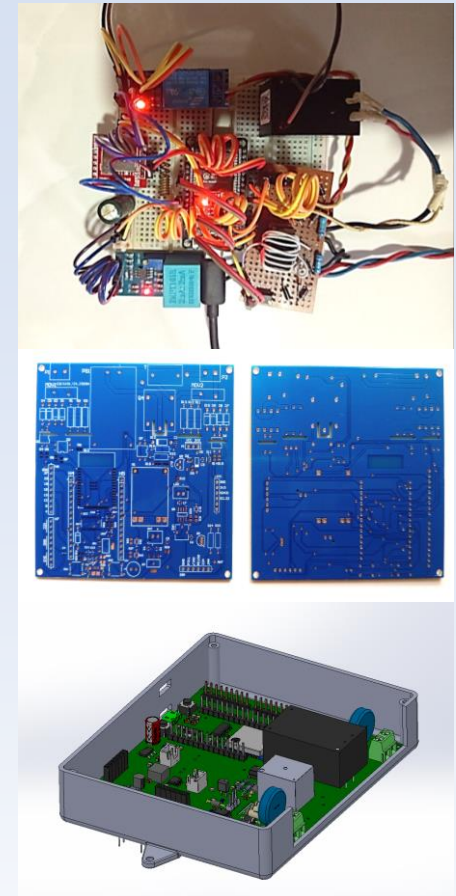
- **ESP32 Microcontroller:** The brain of the system.
- **Optocouplers:** Safely detect AC and DC power
- **Current Transformer:** Checks if current is flowing to the LED screen
- **Battery Backup:** Keeps the system running during power cuts.



Methodology Cont.

Hardware Design (Implementation)

- First, built and tested the entire circuit on a breadboard.
- Then, designed the final circuit board using Altium Designer.
- Finally, designed a protective case in SolidWorks and 3D printed it.





Methodology Cont.

Software & Cloud

- **Firmware:** Coded the ESP32 in C++ for reliable data collection.
- **Communication:** Used **MQTT** protocol to send data.
- **Cloud Database:** Used **InfluxDB** to store all historical data.
- **Backend:** Used **Node.js** to manage the data.

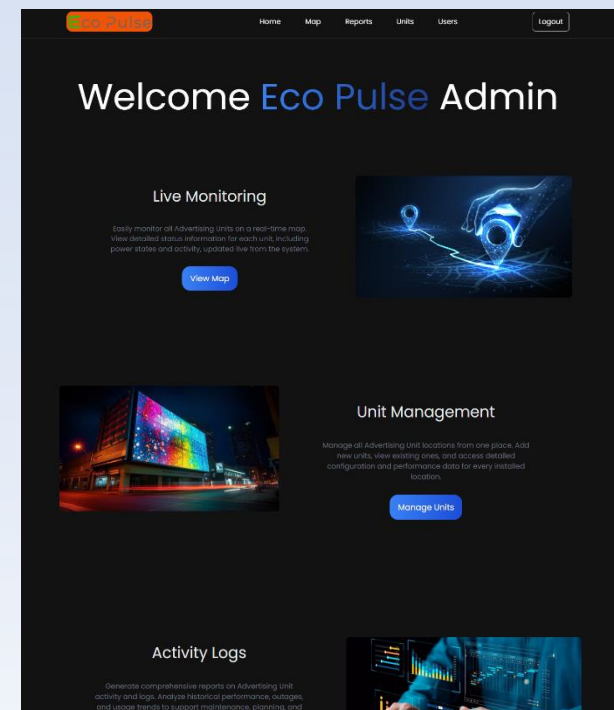




Methodology Cont.

Web Application

- **Platform:** Built a full-stack web application.
- **Frontend:** Developed with **React.js** for a fast, dynamic user interface.
- **Backend:** Used **Node.js** to create the server and API.
- **Database:** **InfluxDB** for storing all time-series device data.

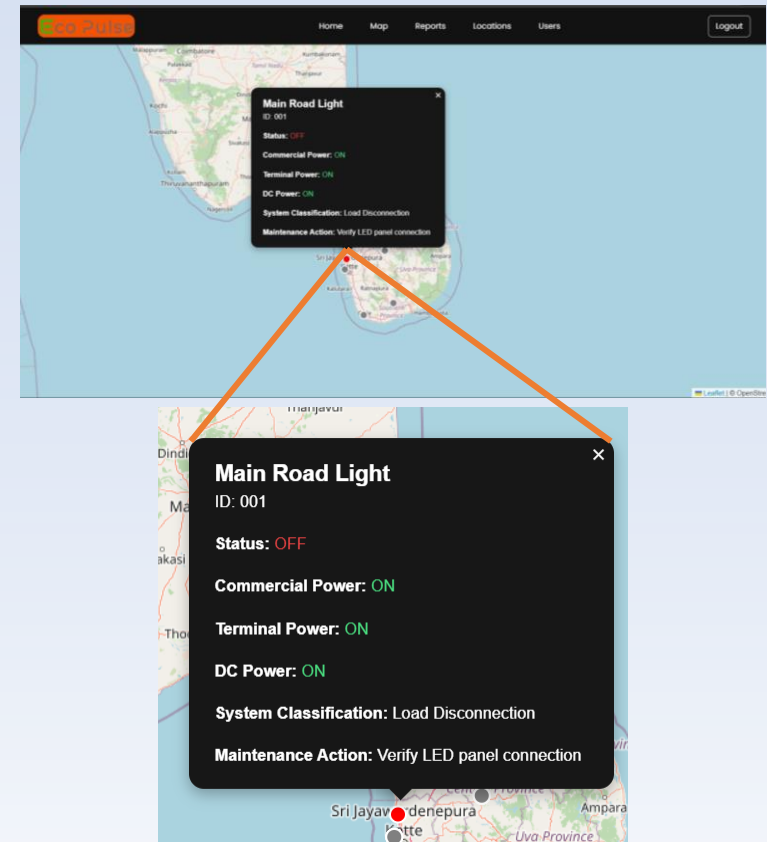




Methodology Cont.

Interactive Map

- **Feature:** Real-time geographical overview of all devices.
- **Technology:** Integrated Leaflet.js with OpenStreetMap.
- **Visual Status:**
 - **Green Marker, Red Marker, Grey Marker**
 - **Functionality:** Click on any marker to see a detailed popup with:
 - Device ID & Location
 - **System Classification:** (e.g., “Load Disconnection”)
 - **Maintenance Action:** (e.g., “Verify LED panel Connection”)

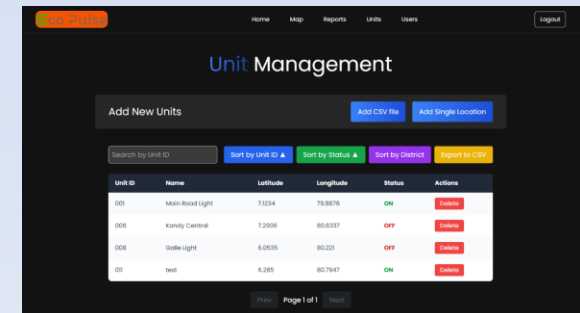




Methodology Cont.

Device & Report Management

- **Device & Location Management:**
 - Add new devices individually or in bulk via CSV upload.
 - Organize devices by location, zone, and district.
 - Edit and manage all device details easily.
- **Automated Report Generation:**
 - Generate detailed reports with one click.
 - Export as PDF (for presentations) and CSV (for data analysis).
 - Filter reports by device, location, and date range.



ADVERTISING MONITORING REPORT

From: 2025-07-03 T06:00:00
To: 2025-07-04 T06:00:00
ID:012
Location Name: Ampara
Location: lat: 7.2912, lng: 81.6725

Stats Table	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-03 T18:00:00
Off	2025-07-04 T18:00:00

Commercial Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00

Device Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00

DC Power	
Stats	Time
On	2025-07-03 T06:00:00
On	2025-07-04 T06:00:00

Summary	
Total Running Time	12.00 hours
Commercial Power	Good
Device Power	Good
DC Power	Good
Error Detections	No



Results

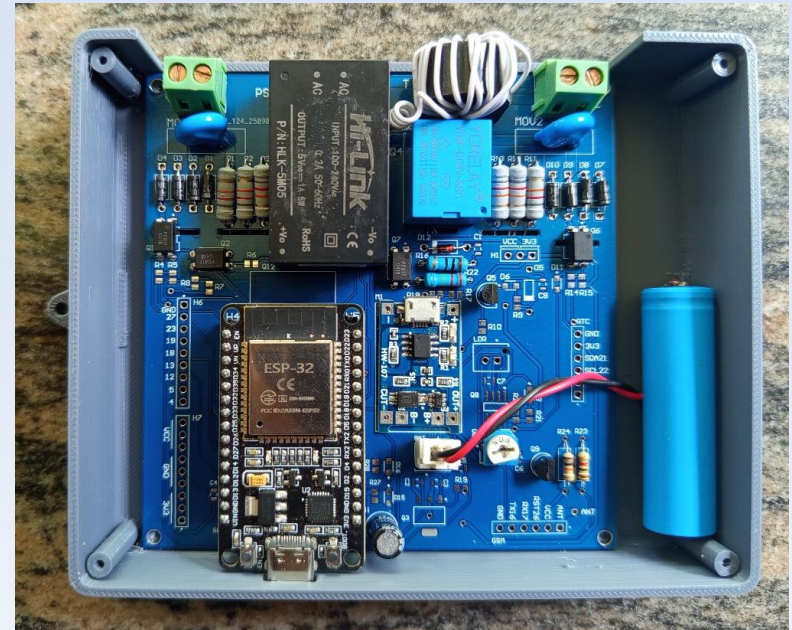
Fault Detection Accuracy

Test Case	Commercial Power	DC Power	Output Terminal	Current Flow	Expected Classification	Actual Result	Pass/Fail
Normal Operation	ON	ON	ON	Present	Normal Operation	Normal Operation	PASS
AC Power Disconnected	OFF	OFF	OFF	Absent	Commercial Power Failure	Commercial Power Failure	PASS
DC Converter Failure	ON	OFF	OFF	Absent	DC Power Supply Failure	DC Power Supply Failure	PASS
Output Terminal Removed	ON	ON	OFF	Absent	Output Terminal Fault	Output Terminal Fault	PASS
Load Disconnected	ON	ON	ON	Absent	Load Disconnection	Load Disconnection	PASS
Relay Switch OFF	ON	ON	OFF	Absent	Output Switching Fault	Output Switching Fault	PASS
System Recovery	ON	ON	ON	Present	Normal Operation	Normal Operation	PASS

Results Cont.

System Performance

- Data Transmission Success: 99%
- Battery Backup: 6.5 hours
- Fast Alerts: Average latency of 25 seconds





Conclusion

- Successfully designed and built a complete, end-to-end IoT monitoring system.
- The innovative Fault Detection solves a key industry problem.
- The system is reliable, accurate, and ready for deployment.
- Provided me with invaluable hands-on experience in embedded systems and full-stack development.



Future Enhancements

- Add Cellular (4G/5G) Connectivity for areas without Wi-Fi.
- Integrate Solar Panels for indefinite battery life.
- Add a Camera/Light Sensor to verify display content.



References

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Thank You